Name_____ Date_____ Period_____

LESSON 7-3 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Absolute Value as Distance

Maryland Standard 6.NS.C.7c · Learning target – I can use absolute value to describe how far a number is from zero.

REMEMBER

- **Absolute value** is a number's distance from 0. We write $|-7| = 7$.
- A number and its opposite have the same absolute value: $|5| = |-5| = 5$.
- Absolute value is never negative. A greater absolute value means **farther from 0**.

WORKED EXAMPLE

Which is farther from 0: −9 or 6?

Step 1 $|-9| = 9$ and $|6| = 6$.

Step 2 $9 > 6$, so −9 is farther from 0.

Answer **−9**

WORD BANK

absolute value – how far a number is from 0

|−5| – the absolute value of −5, which is 5

distance – how far; never negative

PRACTICE

show your work · label units

1. Find each absolute value.

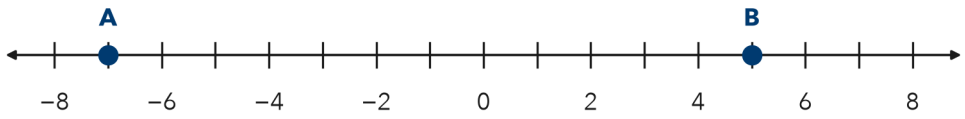
Step 1 Ask: how far is this number from 0?

Step 2 Write that distance. It is never negative.

a. $|-14|$ = _____ **b.** $|8|$ = _____ **c.** $|0|$ = _____

d. $|-2.5|$ = _____ **e.** $|3/4|$ = _____ **f.** $|-1\ 1/2|$ = _____

2. Use the number line. Find the absolute value of each point.



a. $|A|$ = _____ **b.** $|B|$ = _____ **c.** Farther from 0: ____

3. Compare. Write <, >, or = in each blank.

a. $|-9|$ ___ $|7|$ **b.** $|-4|$ ___ $|4|$ **c.** -10 ___ $|-3|$

d. $|-6.5|$ ___ $|6.5|$ **e.** $|-12|$ ___ $|-15|$ **f.** 0 ___ $|-1|$

4. Bank Balances. Rosa's account balance is −$35. She **owes** $35. Ben's balance is $20.

Step 1 Find how far each balance is from 0.

Step 2 The bigger distance is farther from $0.

a. Find the absolute value of each balance. _____

b. Whose balance is farther from $0? _____

5. Stretch. List all the integers with an absolute value less than 3.

Integers_____

6. Reasoning. Name two different numbers that have an absolute value of 4.5. Explain how you know.

Frame *The two numbers are ___ and ___, because both are ___ units from 0.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask someone at home: is it colder at −10 degrees or at 5 degrees? Which temperature is farther from 0?

Name_____ Date_____ Period_____

LESSON 7-3 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Absolute Value as Total Distance

Maryland Standard 6.NS.C.7c · Learning target – I can use absolute value to find the total distance between a negative and a positive reading.

REMEMBER

- **Opposite sides** of 0: **add** the absolute values.
- −3 to 5 is 3 + 5 = 8 units.
- **Same side** of 0: **subtract** the absolute values.
- −9 to −2 is 9 − 2 = 7 units.

WORKED EXAMPLE

A diver swims up from −24 feet to a boat deck at 6 feet. How far does she go?

Step 1 The numbers are on opposite sides of 0, so add.

Step 2 |−24| + |6| = 24 + 6 = 30

Answer **30 feet**

WORD BANK

distance – how far apart two points are

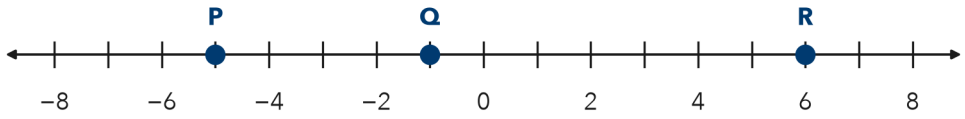
opposite sides of 0 – one negative, one positive: add

same side of 0 – both negative or both positive: subtract

PRACTICE

show your work · label units

1. Use the number line. Find each distance.



a. P to R: _____ **b.** Q to R: _____ **c.** P to Q: _____

2. Find the distance between each pair of numbers.

Step 1 Are the numbers on the same side of 0, or on opposite sides?

Step 2 Opposite sides: add the absolute values. Same side: subtract.

a. −7 and 3 _____ **b.** −12 and −4 _____ **c.** −2.5 and 4 _____

3. **Temperature Swing.** It was −8 degrees at 6 a.m. It was 15 degrees at 3 p.m. How many degrees did it **rise**?

Step 1 From −8 up to 0 is 8 degrees.

Step 2 From 0 up to 15 is 15 degrees. Add.

Answer_____ degrees

4. Elevator. An elevator goes from parking level −3 up to floor 9. Street level is floor 0. How many floors does it travel?

Step 1 From floor −3 up to 0 is 3 floors.

Step 2 From 0 up to floor 9 is 9 floors. Add.

Answer_____ floors

5. Find the Error. To find the distance from −9 to 5, Omar wrote 9 − 5 = 4. Explain his mistake. Find the correct distance.

Frame *Omar is wrong because −9 and 5 are on ___ sides of 0, so he should ___. The distance is ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. What is the distance between −13 and 6 on a number line?

A 7 **B** 19 **C** −19 **D** −7

7. A cave floor is at −60 feet. A ledge is at −22 feet. How far apart are they?

A 82 feet **B** 38 feet **C** −38 feet **D** 22 feet

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME With someone at home, find how many degrees the temperature changed today.